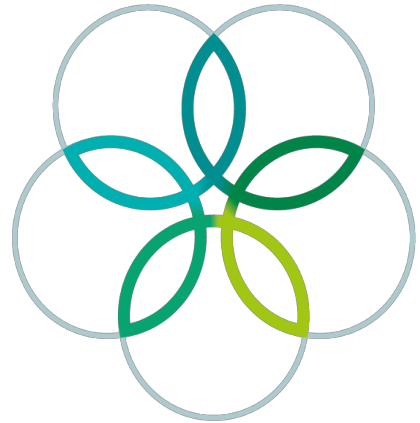
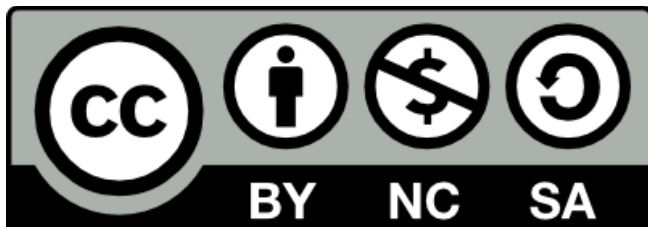


INTERNATIONAL  
BIOLOGY  
OLYMPIAD e. V.

IBO



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## V INTERNATIONAL BIOLOGY OLYMPIAD

1. A scientist made several sections of a living cell at different levels. After viewing these sections under an electron microscope, he drew a complex diagram as shown on the figure. It would be reasonable to conclude that this cell is **not**:

- a) photosynthetic;
- b) respiratory;
- c) eucaryotic;
- d) a plant cell.

2. The two types of cell organelles that transform energy are:

- a) chromoplasts and leucoplasts
- b) mitochondria and leucoplasts;
- c) mitochondria and chloroplasts
- d) mitochondria and chromoplasts.

3. A centriol is:

- a) stable structure in the cytoplasm near the nucleus;
- b) stable structure within the nucleus;
- c) structure, appearing during the mitosis;
- d) structure, which is a part of the chromosome.

4. How is it possible that cells of one and the same kind can contain different number of plasmids:

- a) it is because of the circular structure of plasmids;
- b) because plasmids are able to replicate independently of chromosomes;
- c) it depends on the molecular weight of plasmids;
- d) because plasmids are usually smaller than chromosomes.

5. In eucaryotic cells DNA may be found in the:

- a) nuclei
- b) mitochondria
- c) chloroplasts
- d) all of these.

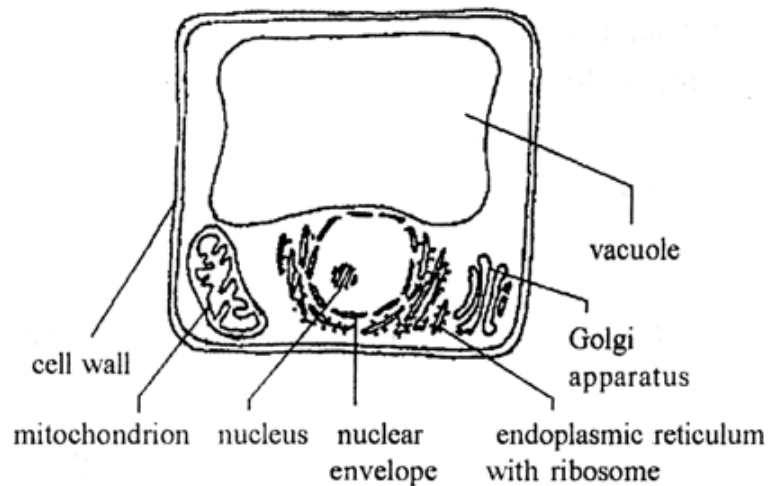
6. A poison that interferes with the synthesis of proteins is most likely to interfere with the function of the:

- a) cytoplasm
- b) centromeres
- c) ribosomes
- d) vacuoles.

7. Which one of the following characteristics does not correspond to the properties of water?

- a) bad heat conductor;
- b) bad electrical insulator;
- c) good solvent for majority of substances in cells;
- d) it takes part in majority of chemical reactions in cells.

8. According to the fluid-mosaic model of the cell membrane:

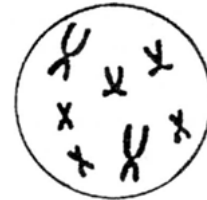


- a) proteins and phospholipids form a regular repeating structure;
- b) the membrane is a rigid structure;
- c) phospholipids form a double layer, with the polar parts facing each other;
- d) proteins are fairly free to move laterally within a double layer of phospholipids.

**9. Which one of the following statements is not correct. Active transport of a substance:**

- a) can occur against the concentration gradient;
- b) can occur against the electrochemical gradient;
- c) can be slowed by a factor of at least two by lowering the temperature by 10°C;
- d) is unaffected by metabolic poisons;
- e) shows saturation properties.

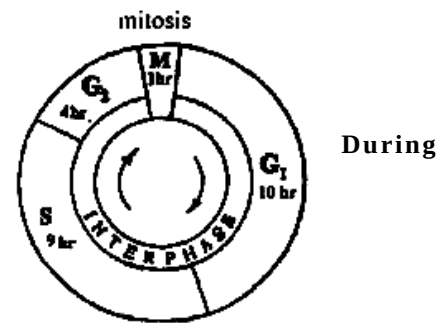
**10. An abnormal specimen of some animal species was produced by sexual reproduction. The figure below shows the chromosomes of a somatic cell of this organism. One of the gametes that contributed to this specimen had a wrong number of chromosomes. How many chromosomes were contained in this gamete?**



- a) 4
- b) 5
- c) 6
- d) 7.

**11. This question refers to the following diagram. the S-phase of cell cycle the cell:**

- a) undergoes cytokinesis;
- b) undergoes meiosis;
- c) replicates its DNA;
- d) undergoes mitosis;
- e) enters interphase.



**12. The chromosomes number in tomato plants is 24. A tomato cell undergoes meiosis. Three of the resulting cells degenerate. The last cell immediately undergoes three mitoses without cytokinesis. How many cell nuclei will you find; how many chromosomes does each of these nuclei contain:**

- a) 4 nuclei with 12 chromosomes each;
- b) 4 nuclei with 24 chromosomes each;
- c) 8 nuclei with 12 chromosomes each;
- d) 8 nuclei with 24 chromosomes each.

**13. Repair regeneration means:**

- a) general property of the tissues and organs to renew throughout cell life and aging;
- b) physiological changes that take place during the period of differentiation and specialization of cells;
- c) restoration of tissues and organs after induced injury;
- d) replacing of old cells with new ones.

**14. The unique properties of each amino acid are determined by its particular:**

- a) R-group
- b) amino group
- c) type of the peptide bonds;
- d) number of the bonds with other amino acids.

**15. A mutation that changes an alanine residue in the interior of a protein molecule to valine is found to lead to a loss of activity. The reason is:**

- a) valine is acidic but alanine is basic;
- b) alanine is necessary for  $\alpha$ -helix formation, whereas valine is not;
- c) valine occupies more space than alanine, hence the shape of the protein molecule is changed;
- d) the presence of valine changes the isoelectric point of the protein.

**16. Poly-L-Leucine in an organic solvent is  $\alpha$ -helical, whereas poly-L-isoleucine is not. The reason for this behaviour of poly-L-isoleucine is:**

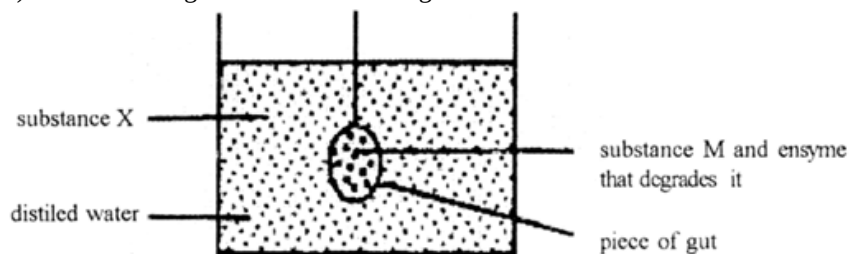
- a) the methyl group attached to the  $\beta$ -carbon of isoleucine sterically interferes with the  $\alpha$ -helix formation;
- b) the peptide bonds in poly-L-leucine are different;
- c) isoleucine is less soluble in organic solvents than leucine;
- d) the isoelectric point of isoleucine is higher.

**17. The enzyme succinic dehydrogenase normally catalyzes a reaction involving succinic acid. Another substance, malonic acid, sufficiently resembles the succinic acid to forming temporary complexes with the enzyme, although malonic acid itself cannot be catalyzed by the succinic dehydrogenase. In this example succinic acid and malonic acid are, respectively:**

- a) a substrate, a competitive inhibitor;
- b) a competitive inhibitor, a substrate;
- c) a positive modulator, a negative modulator;
- d) a negative modulator, a positive modulator.

**18. A piece of gut which contains a substance M and an enzyme that degrades this substance in it is immersed in a container with distilled water. When the water is examined on the next day it is observed that it contains only a substance X. Based on this observation which one of the following statements cannot be speculated:**

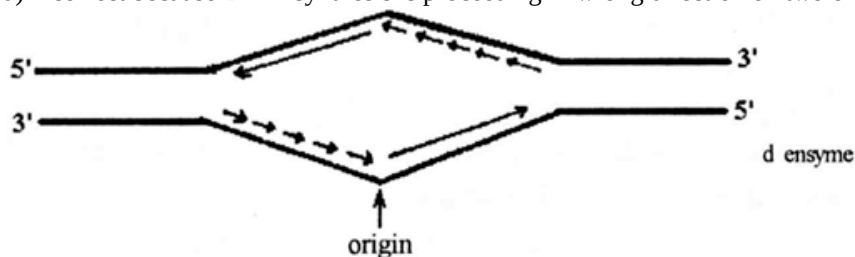
- a) substance X is degraded by the enzyme M;
- b) substance M is degraded in the gut;
- c) substance X is the constituent of substance M;
- d) substance X can penetrate the gut wall;
- e) molecular weight of substance M is greater than that of substance X.



**19. The diagram below represents a student's view of DNA synthesis occurring in an animal cell. The arrows represent newly synthesized DNA.**

**This diagram is:**

- a) correct as shown;
- b) incorrect because DNA synthesis in animal cells is unidirectional;
- c) incorrect because DNA synthesis proceeds in 3' - 5' direction;
- d) incorrect because DNA synthesis is proceeding in wrong direction on two of the strands.



**20. A synthetic mRNA has only two nucleotide types; it contains five times as much U as C:**

- a) there are 4 possible codons
- b) there are 6 possible codons;
- c) there are 8 possible codons
- d) there are 16 possible codons.

**21. The exact replication of DNA is possible due to:**

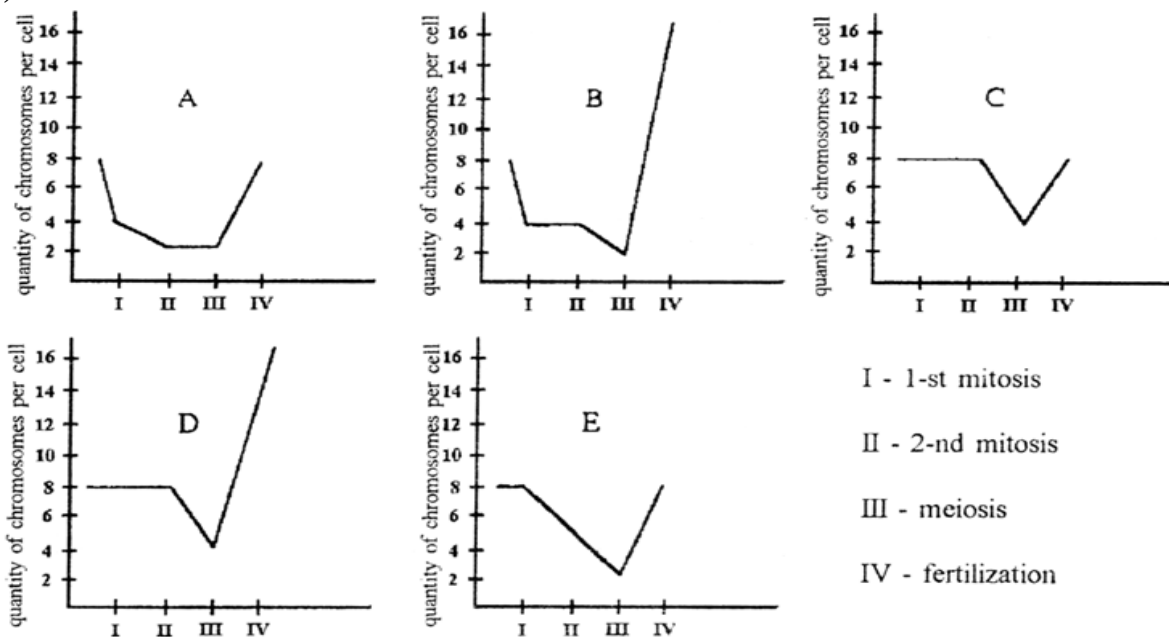
- a) the genetic code
- b) mitosis
- c) the base-pairing rules;
- d) the fact that the **DNA** molecules are enclosed within a nuclear membrane.

**22. Which one of the following statements explains why penicillin does work as an antibiotic:**

- a) penicillin inhibits the nucleic acid synthesis;
- b) penicillin inhibits murein synthesis, the material necessary for building of bacterial wall;
- c) penicillin inhibits protein synthesis among procaryotes;
- d) eucaryotes rapidly decompose penicillin.

**23. A cell with eight chromosomes undergoes two mitoses and one meiosis. One of the daughter cells is fertilized. Which one of the following figures represents the changes in the chromosome number of the above mentioned cell.**

- a) A;
- b) B;
- c) C;
- d) **D**;
- e) E.



**24. Which one the following statements refers to a human cell with 22+X chromosomes?**

- a) it is a fertilized egg
- b) it is a cell which has undergone a meiosis;
- c) it is a somatic cell
- d) it is a cell which has undergone a mitosis;
- e) it is a polyploid cell.

**25. Which one of the following substances could be subjected to denaturation?**

- a) only proteins
- b) only nucleic acids:
- c) only lipids
- d) nucleic acids and proteins.

**26. Which one of the following statements is correct? Denaturation is a process of:**

- a) losing of the secondary and tertiary structure and destroying the primary structure;
- b) losing of the secondary and tertiary structure without destroying the primary structure;
- c) only losing of quaternary structure;
- d) reducing of the chemical reactivity of the polymers.

**27. The Krebs cycle takes place in the:**

- a) mitochondria;
- b) cytoplasm;
- c) chloroplasts;
- d) nucleus.

**28. The Krebs cycle is a source of:**

- a) only ATP
- b) nucleoside triphosphates and important intermediate metabolites;
- c) only glucose
- d) only pyruvic acid.

**29. Which of the following substances are products of aerobic respiration:**

- a) only  $\text{CO}_2$  and  $\text{H}_2\text{O}$
- b)  $\text{CO}_2$ ,  $\text{H}_2\text{O}$  and ATP
- c) oxalic acid
- d) citric acid.

**30. Which one of the following statements is not correct? Allosteric affect on the enzyme activity is a way for:**

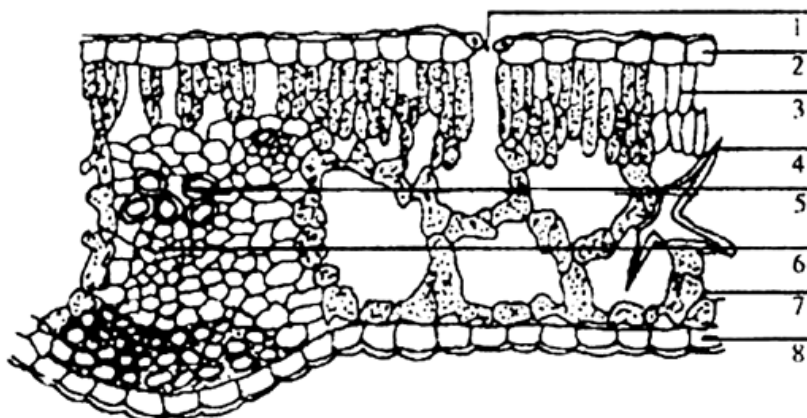
- a) regulation of metabolic processes
- b) activation of enzyme activity;
- c) competitive inhibition of enzyme activity;
- d) noncompetitive inhibition of enzyme activity.

**31. The main functions of the glycocalyx are:**

- a) structural and protective;
- b) mechanosupporting and protective;
- c) maintaining the environment immediately around the cell, fulfilling of the selective permeability of compounds necessary for the cell and extracellular digestion;
- d) cell contaction and carrying out of the intercellular exchange.

**32. This figure is a cross section at one kind of leaf. What is the type of tissue (cell) of number 4?**

- a) palisade cell;
- b) parenchyma;
- c) sponge cell;
- d) sclereid.



**33. Cells of xylem vascular**

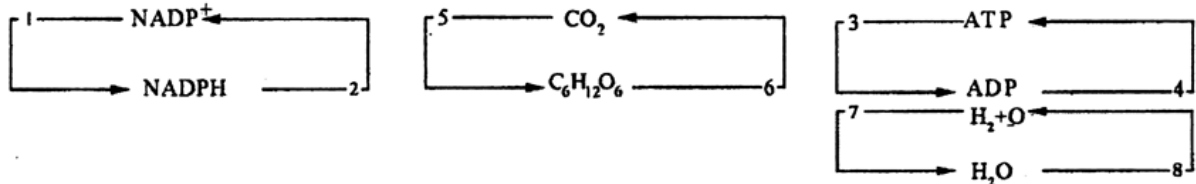
**bundles are in the period of their activity:**

- a) dead
- b) alive, only their cell membranes become woody;
- c) alive, only their nucleolus disappears;
- d) alive, with a big vacuole inside of the cell.

34. Some blue-green algae (*Cyanobacteriae*) are both autotrophic (photosynthetic) and heterotrophic (being able to assimilate organic compounds). It has been shown by experiments that light promotes the assimilation of organic compounds by them. Which one of the following is the most probable explanation of this phenomenon?

- the presence of organic compounds inhibits photosynthesis;
- the uptake and utilization of organic compounds by blue-green algae consume no energy;
- ATP and NADPH produced in photosynthesis can be used for the uptake and further transformation of organic compounds;
- blue-green algae can fix atmospheric nitrogen.

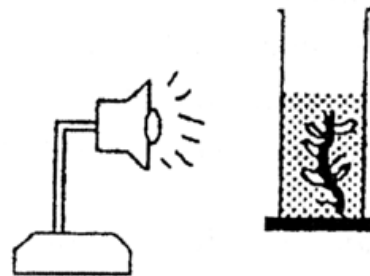
35. Which one of the following processes constitute a part of the light- dependent photosynthetic reactions?



- 1, 3, 6;
- 1, 4, 8
- 2, 3, 6;
- 2, 4, 5;
- 5, 7.

36. Which of the following does not affect the number of gas bubbles per unit of the time in the above figure?

- volume of the glass container;
- intensity of the light;
- number of the leaves on the plant;
- amount of CO<sub>2</sub> dissolved in the water;
- temperature of the water.



37. In C<sub>3</sub>-photosynthesis the CO<sub>2</sub> acceptor is:

- 3-phosphoglycerine aldehyde
- pyruvic acid;
- ribulose-1,5-diphosphate
- ferredoxin.

38. In photosynthesis, in contrast to mitochondrial respiration:

- water is decomposed
- water is formed;
- electrons are transported with the aid of cytochromes, among others;
- participation of colorful metallocompounds;
- more than one of the above are characteristic for photosynthesis only.

39. In the 1930th, C.B. Van Niel correctly hypothesized that oxygen gas released by plants come from:

- H<sub>2</sub>O
- CO<sub>2</sub>
- C<sub>6</sub>H<sub>12</sub>O<sub>6</sub>
- O<sub>3</sub>;

40. Auxins function as:

- stimulators of leaves and fruits drop;
- stimulators of lateral bud growth;
- inhibitors of length growth;
- stimulators of cell elongation and root development.

**41. Besides the three factors: photoperiod, temperature and nutrition, another factor, which also controls the flowering of plants is:**

- a) the rate of photophosphorylation;
- b) the rate of the transport of organic substances within the plant;
- c) the age of the plant;
- d) the pH of soil solution.

**42. Plant cells were incubated for several hours in the presence of radioactive (tritium labeled) uridine. Afterwards cells were collected, gently homogenized and fractionated to obtain various organelles. Radioactivity will be found chiefly in:**

- a) nucleolus, plastids and Golgi apparatus;
- b) nucleus, nucleolus and lysosomes;
- c) nucleus, ribosomes and vacuoles;
- d) nucleus, nucleolus, ribosomes and chloroplasts;
- e) all of the mentioned organelles will exhibit roughly the same level of radioactivity.

**43. The number of seeds in a pod is determined by the number of:**

- a) pistils in the flower
- b) carpels in the pistil;
- c) ovules in the ovary
- d) embryos in the ovular sac;
- e) stigmata in the pistil.

**44. Which of the following refers to the sporophyte in plants:**

- a) diploid that produces haploid cells by mitosis;
- b) diploid that produces haploid cells by meiosis;
- c) haploid that produces gametes by meiosis;
- d) haploid that produces gametes by mitosis.

**45. When the root cap of a maize seedling is excised, a new root cap will be formed within about 5 days. But in a flying spaceship, there is no regeneration of the root cap. What is the reason?**

- a) root cap formation is related to the CO<sub>2</sub> content of the atmosphere;
- b) root cap formation is related to photoperiod;
- c) root cap formation is related to temperature difference between day and night;
- d) root cap formation is related to gravity.

**46. In healthy adult humans, red blood cells are formed only in:**

- a) sternum, ribs and vertebrate bodies
- b) thigh and shank bones;
- c) liver;
- d) spleen.

**47. As blood flows through the heart the valves open and close at the correct moments. These valves are forced to open by:**

- a) the contraction of the powerful muscles of the ventricles;
- b) small muscles in the valves themselves;
- c) strings attached to them called *chordae tendineae*;
- d) the pressure of the blood itself.

**48. Which of the following statements explains the persisting abnormally high level of glucose in the human blood:**

- a) insufficient filtration by the kidneys;
- b) excessive amount of thyroid hormones in the blood;
- c) muscles in a total resting state (motor hypoactivity, hypodynamics);
- d) insufficient hormone secretion by the pancreas;
- e) an excessive amount of glucose absorbed by the intestine.

**49. Why the human organism cannot be fed only by carbohydrates is:**

- a) because the human body is composed mainly of proteins;
- b) because man for his life needs many other substances, that cannot produce himself;
- c) because the calorific value of the carbohydrates is not very high;
- d) because the carbohydrates do not contain nitrogen and the human organism cannot assimilate it from the air.

**50. A student added saliva to starch solution and expected the starch to be converted into mono- and disaccharides. Which one of the following is the best condition for his experiment?**

- a) the mixture was kept at 0°C;
- b) the mixture was kept at 30°C;
- c) the mixture was boiled and then kept at 30°C;
- d) the mixture was kept at 70°C.

**51. Basal metabolic rate is:**

- a) the minimal amount of energy needed to maintain life under exactly defined conditions;
- b) issue of energy after obtaining nutrition - 100 g of proteins;
- c) issue of energy by effort - ten knee-bends;
- d) amount of energy needed for converting 1 g of lipids.

**52. Mammalian species X and Y have similar overall shapes and body cover and exhibit similar locomotory activity. However, individuals of species X are twice as large as those of species Y. The loss of heat through their surface, per unit of body mass, for these species will be about:**

- a) for X twice as large as for Y
- b) for X four times as large as for Y;
- c) for Y twice as large as for X
- d) neither of the given answers is correct.

**53. A muscle has built up an oxygen debt. When there is enough oxygen for aerobic respiration to resume, all of the following will occur except:**

- a) lactic acid will be converted into pyruvic acid;
- b)  $O_2$  will be used up;
- c) acetyl-CoA will be converted into  $CO_2$  and  $H_2O$ ;
- d) pyruvic acid will be converted into acetyl-CoA;
- e) an excess of NADPH will accumulate.

**54. Excessive concentration of thyroxine causes uncoupling of mitochondria, i.e. in spite of the electrons flowing along the oxidative chain no ATP is produced. Based on this fact one should expect people with excessive activity of the thyroid to be:**

- a) obese in spite of consuming small amounts of food and with body surface colder than in healthy individuals;
- b) obese in spite of consuming small amounts of food and with body surface warmer than in healthy individuals;
- c) lean in spite of consuming large amounts of food and with body surface colder than in healthy individuals;
- d) lean in spite of consuming large amounts of food and with body surface warmer than in healthy individuals.

**55. Blood hydrostatic pressure powers the process of:**

- a) filtration across the glomerulus;
- b) reabsorption of water and dissolved substances at the venule end of capillary;
- c) sodium gradient maintenance in the kidney tubules;
- d) salt and glucose reabsorption in the kidney tubules;
- e)  $Na^+$  reabsorption in the loop of Henle.

**56. In the mammalian embryo, the first structure to develop relationship with the wall of the uterus is?**

- a) the amnion;
- b) the chorion
- c) the allantois
- d) the trophoblast.

**57. Which of the following is a common function of the adrenal glands, the pancreas and the liver in human?**

- a) regulation of the metabolic rate
- b) construction of the blood vessels;
- c) synthesis of urea and uric acid
- d) digestion of fats;
- e) taking part in the regulation of glucose level in the blood.

**58. The structure of hormones cannot be:**

- a) amino acid derivatives
- b) peptides and proteins;
- c) steroids
- d) carbohydrates.

**59. In an individual with short-sightedness (myopia):**

- a) the eye tends to be longer than average from lens to retina;
- b) use of appropriate biconvex lens corrects the defect;
- c) close vision is more seriously affected than distant vision;
- d) a circular object tends to appear oval.

**60. A drug that inactivates acetylcholinesterase:**

- a) inhibits the release of acetylcholine from presynaptic endings;
- b) inhibits the attachment of acetylcholine to its receptor protein;
- c) increases the duration of acetylcholine stimulating muscle contraction;
- d) all of above.

**61. All the following diseases are caused by a protozoan except:**

- a) malaria
- b) sleeping sickness
- c) rabies
- d) amebic dysentery.

**62. Which one of the following statements referring to the axon is not correct?**

- a) the axon is the long projection of the neuron;
- b) the myelin sheath accelerates the nerve impulse conduction;
- c) the axons could be longer than two meters;
- d) some axons have not myelin sheath;
- e) the speed of nerve impulse conduction does not depend on the axon diameter.

**63. Hemopoietic organs are:**

**The codes of the answers: 1. marrow 2. thymus 3. islets of Langerhans 4. spleen; 5. lymph nodes; 6. core of the adrenal glands. Answers:**

- a) 1, 2, 3 and 4;
- b) 3, 4 and 6;
- c) 1, 2, 4 and 6;
- d) 1, 2, 5 and 6;
- e) no correct answer.

**64. To reach the right hand the blood from the stomach and intestine:**

**The codes of the answers: 1. must pass through the heart (once) 2. must pass through the heart (twice) 3. do not pass through the heart 4. must pass through the lungs; 5. must pass through the liver; 6. must pass through the brain. Answers:**

- a) only 2
- b) 1 and 4
- c) 2, 4 and 6
- d) 2, 4 and 5; e) 3, 4 and 5.

**65. A hungry brown rat was placed in a closed metal box with a food slot and a food-releasing lever. The rat was allowed to poke around randomly until it accidentally tripped the lever and released a food pellet. The rat soon learned to trip the lever whenever it wanted food. What type of learning is this:**

- a) imprinting;
- b) sensitization
- c) habituation
- d) trial and error.

**66. Which process of cell division is essential if Mendel's First Law (the Law of Segregation) is to be fulfilled:**

- a) division of the centromere
- b) duplication of chromosomes;
- c) pairing of homologous chromosomes
- d) formation of chiasmata.

**67. Man A has a rare X-linked recessive factor which is expressed as trait A. Man B has a rare dominant autosomal factor which is expressed only in males as trait B. Would you distinguish these two cases just by studying the progeny:**

- a) no, because the progeny of man A and man B can have just the same trait of their respective male ancestors;
- b) yes, because in the progeny of man A and man B would not be woman with trait B;
- c) no, because man A would not have sons with trait A, and man B would not have sons with trait B;
- d) yes, because the sons of man A's daughter could be with trait A, whereas the sons of man B's daughter and the sons of his son could both be with trait B.

**68. In numerous bird species males have very colorful and ornate plumage.**

**Choose the best explanation for the evolutionary stabilization of such a feature:**

- a) ornate plumage is a good camouflage among colorful tropical plants;
- b) this feature is neither advantageous, nor detrimental, so is neither selected for, nor against;
- c) females mate preferentially with ornate males, so the genes "causing" this feature spread among the population;
- d) ornate plumage makes it easier for the birds of the same species to recognize each other, thus potentially dangerous encounters with individuals of other species are being avoided;
- e) bright and ornate plumage serves as a warning against dangerous (often poisonous) species.

**69. Very similar species of fish release their eggs and sperm into the same water, but the sperm of one species cannot penetrate the eggs of the other species. This is an example of prezygotic isolation by:**

- a) ecological isolation
- b) behavioral isolation;
- c) mechanical isolation
- d) gametic isolation.

**70. In silk worms, silk gland cells are specialized in the synthesis of large quantities of the protein known as silk-fibroin. These same cells do not make blood specific proteins. One would expect that silk-gland cells have:**

- a) only silk fibroin genes;
- b) the genes for both blood protein and silk fibroin;
- c) silk fibroin genes and some other genes, but not blood protein genes;
- d) fewer genes than the zygote.

**71. Pleiotropic gene activity could be revealed mainly through:**

- a) studying of genes;
- b) studying the interaction of genes of the organism;
- c) studying pedigrees (genealogy);
- d) studying of phenotypic changes, caused by gene mutations.

**72. Which of the following conditions have enabled the appearance of the living organisms on dry land:**

- a) decreasing of the vulcanic activity and tectonic processes;
- b) decreasing of the sea level and enlarging of the land surface;
- c) formation of the ozone layer in the atmosphere;
- d) general warming of the climate, increasing of humidity, appearing of the green-house effect.

**73. Comparing the skeleton of man and other primates the following peculiarities could be determined:**

**The codes of the answers:**

**1. The thorax of man is laterally flattened and that of the primates is almost cylindrical, with massive bones to which strong muscles are attached.**

**2. The bones of the foot of man are shorter.**

**3. In man the aperture of the lower part of the skull, serving for the connection between the brain and the spinal cord is situated backwards, while in the primates it is driven more forward.**

**4. The molars of man are with U-shaped surface and these of the primates are with X-shaped surface.**

**Answers:**

- a) only 1 and 3
- b) only 1, 2 and 3
- c) only 2 and 3;
- d) only 2 and 4 e) all answers are correct.

**74. In a family the father has blood group A and his son has blood group 1 and is a hemophylic. What is the possible genotype of the mother, the son and the father in that family?**

**The codes of the answers:**

| <i>father</i>        | <i>mother</i>       | <i>son</i>        |
|----------------------|---------------------|-------------------|
| 1. $I^A I^O X^H Y$ , | $I^B I^B X^H X^H$ , | $I^B I^O X^h Y$ ; |
| 2. $I^A I^O X^H Y$ , | $I^B I^O X^H X^h$ , | $I^B I^O X^H Y$ ; |
| 3. $I^A I^O X^h Y$ , | $I^A I^B X^H X^h$   | $I^B I^O X^h Y$ ; |
| 4. $I^A I^A X^h Y$ , | $I^B I^O X^H X^h$   | $I^B I^O X^H Y$   |

**Answers:**

- a) only 1 and 3
- b) only 2 and 4
- c) only 1 and 4;
- d) only 2 and 3 e) only 2, 3 and 4.

**75. What is the most probable sequence of the genes ABCD if the distance between them is: A-B = 1,5 map units; B-C = 18 map units and A-D = 18,5 map units.**

**The codes of the answers:**

- 1. ABCD; 2. BACD;
- 3. ABDC; 4. ACBD.

**Answers:**

- a) only 1
- b) only 3;
- c) 2 or 3
- d) only 4;
- e) all answers are incorrect.

76. After crossing a guinea-pig with black hair and guinea-pig with white hair, all  $F_1$  individuals were with black hair. After analyzing the progeny of a  $F_1$ - female crossed with a homozygous recessive individual the phenotypes in the backcross were in the following ratio 3 white : 1 black. What are the possible genotypes of the parents and individuals of  $F_1$ ?

The codes of the answers:

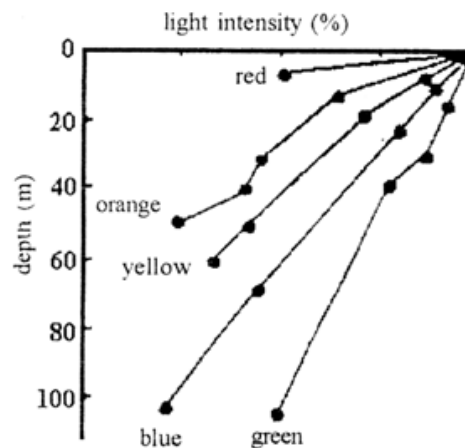
1. AA, aa, Aa;
2. AAbb, aaBB, AaBb;
3. AABB, AAbb, AABb;
4. AABB, aabb, AaBb.

Answers:

- a) only 1;
- b) only 3;
- c) only 2;
- d) only 4
- e) all answers are incorrect.

77. The graphic shown indicates the amount and color of light penetration in the sea. Considering this data, one would expect the basic mass of phytoplankton to be situated in the level:

- a) 1-5 m
- b) 5-10 m
- c) 10-20 m
- d) 20-30 m
- e) 30-40 m.



78. A plant, living in a hot, dry environment is likely to have:

- a) large, dark green leaves with many stomata on both sides;
- b) large, light green leaves with stomata on one side only;
- c) medium, light-colored hairy leaves with stomata mostly on the underside;
- d) small, thick leaves with few stomata.

79. An epiphyte is a growth form adapted for gaining access to:

- a) light
- b) pollinators;
- c) water
- d) nutrients.

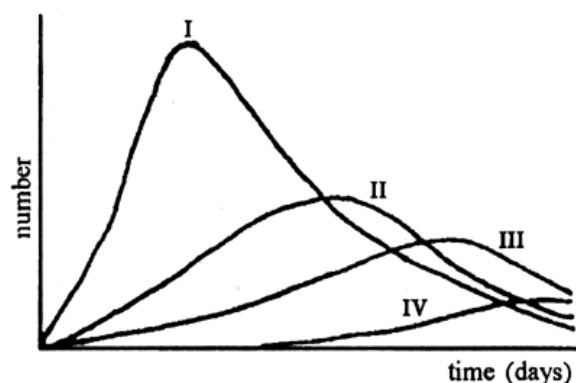
80. The highest osmotic pressure is characteristic for the cells of:

- a) hydrophytes
- b) mesophytes;
- c) xerophytes
- d) halophytes.

81. A class of students inoculated a flask of leaf broth with some pond mud and kept in darkness. Samples were taken daily to ascertain the abundance of microorganisms in the broth over a ten-day period. The results are shown.

Which one of these statements is not a valid conclusion:

- a) organism I is probably a producer;
- b) organism II probably eats organism I;
- c) organism II succeeds organism I;
- d) organism IV is probably a predator.



82. The water-hyacinth was introduced into a tropical country and has become a major menace to boating, clogging rivers and lakes. A single plant can produce 1000 offsprings in 50 days.

The reason for its success is:

- a) lack of predators;
- b) good climate;
- c) high reproductive rate;
- d) the empty niche it occupied;
- e) all of the above answers could be correct.



83. Plants with soft stems and badly developed supportive tissue, few stem parenchyma cells with large intercellular spaces and with a thin cuticle, belong to which ecological group?

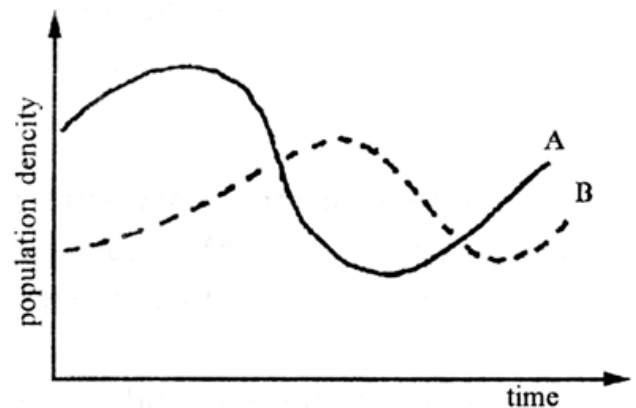
- a) mesophytes
- b) xerophytes
- c) hydrophytes
- d) halophytes.

84. Plants that have mutualistic relations with nitrogen fixing bacteria provide the bacteria with:

- a)  $N_2$
- b) enzymes;
- c) sugars;
- d) nitrites.

85. The figure represents the changes in the density of population of species A and species B. What is the most possible relationship between species A and B?

- a) parasite - host;
- b) predator - prey;
- c) competition;
- d) symbiosis.

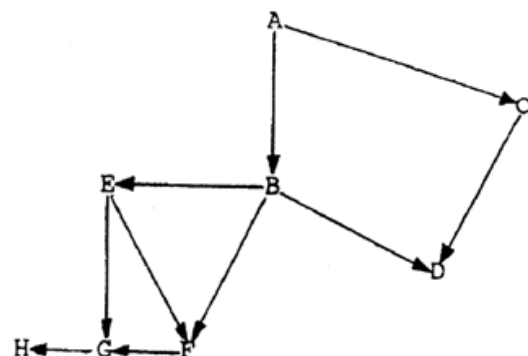


86. Organisms belonging to two closely related animal species were placed in a container with all parameters kept constant, and a limiting supply of food. There is no direct interspecies interaction. In the state of equilibrium:

- a) there will be constant numbers of animals of both species (static equilibrium), irrespectively of the initial numbers;
- b) there will be oscillating numbers of animals of both species (dynamic equilibrium);
- c) there will be constant numbers of animals of both species (static equilibrium), determined by the initial numbers;
- d) animals of only one species will survive;
- e) each of the above situations may take place, depending on the species used for the experiment.

87. If the above ecosystem represented by the food webs above was sprayed with DDT, the greatest concentration of this insecticide would be expected in:

- a) A;
- b) B
- c) G
- d) F.



**88. A potential danger to an isolated population that has been greatly reduced in number is the:**

- a) loss of genetic variability
- b) tendency toward assortative mating;
- c) reduced gene flow
- d) Hardy-Weinberg disequilibrium.

**89. The struggle for existence is a consequence of:**

- a) each organism leaving more offspring than needed to replace itself;
- b) innate competitive tendencies;
- c) the inevitable difficulty of coping with climatic conditions;
- d) territories and dominance hierarchy.

**90. In a stable ecosystem of a pond sudden heavy pollution causes death of all plants. The first visible change in pond water composition will be decreased in the concentration of:**

- a) carbon dioxide
- b) nitrates
- c) oxygen;
- d) phosphates e) answers a and c are correct.

**91. Which of the following promotes diversity of living things:**

- a) classification of organisms;
- b) natural selection in different environments;
- c) inheritance from a common ancestor;
- d) homeostatic regulation.

**92. Which of the following cannot be a result of the ecological succession:**

- a) increase in the trophic levels in the food chain;
- b) increase in the productivity;
- c) increase in the stability of the community;
- d) gradual decrease in the organic material (humus);
- e) increase in the species diversity.

**93. Study the pictures and the above key. Which picture fits key 2a?**

**Key:**

**1a. single flower;**

**1b. inflorescence.**

**2a. pistil flower and perfect flower;**

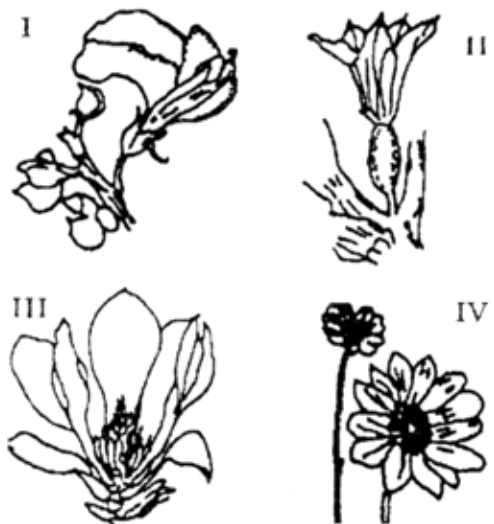
**2b. perfect flower only.**

**3a. one ovary in a flower;**

**3b. many ovaries in a flower.**

Answers:

- a) I
- b) II c) III
- d) IV.



**94. Secondary growth of the stem is typical for:**

- a) *Bryophyta*, *Gymnospermae*, *Angiospermae*;
- b) *Angiospermae*; c) *Dicotyledones*;
- d) *Gymnospermae*, *Dicotyledones*.

**95. The most diverse and widespread of all contemporary plants are:**

- a) mosses
- b) ferns
- c) conifers
- d) angiosperms.

**96. In what way is the growth of a flowering plant most different from the growth of a mammal:**

- a) parts of the organism grow at different rates;
- b) the growth at one part might be controlled by another part;

- c) the total mass of the organism remains relatively constant once maturity is reached;  
d) growth tends to become confined to special groups of cells.

**97. Which of the following structures is common in fish, bird and mammalian auditory (hearing) organ:**

- a) auriculum  
b) semicircular channels;  
c) auditory (hearing) channel  
d) Eustachian tube;  
e) a well developed cochlea.

**98. Choose the correct statement:**

- a) reptiles like fish and amphibians have a body temperature regulating mechanism;  
b) the bodies of reptiles are covered by moist, soft scales that help to protect them against dehydration;  
c) due to the development of kidney tubules as water conserving structures, reptiles are able to survive in the desert;  
d) reptiles secrete nitrogenous waste as urea.

**99. Of the following groups the first animals to serve as pollinators were:**

- a) ant  
b) beetle  
c) bees  
d) hummingbird.

**100. The sense organ of balance in the crayfish is the:**

- a) claws  
b) ears  
c) antennal statocyst;  
d) first mouth appendages; e) eye stalks.

**101. Heart with two atria and one ventricle can be found in:**

- a) sparrow  
b) toad  
c) ray  
d) carp.

**102. Which animal is reproduced by parthenogenesis as a normal process:**

- a) hydra;  
b) tapeworm  
c) earthworm;  
d) honey bee.

**103. Approximately what percentage of existing animal species are invertebrates:**

- a) 20 %  
b) 50 %  
c) 70 %  
d) 95 %.

**104. Study the following diagrams of cell division. For "stage of cell division" choose one alternative from the list below - write the figures in the table.**



**Fig.1**



**Fig.2**

|                                          | Fig. 1 (104.1) | Fig. 2 (104.2) |
|------------------------------------------|----------------|----------------|
| Stage of cell division                   |                |                |
| Number of 2n chromosomes (in interphase) |                |                |

**The codes of the answers:**

1. mitotic prophase    5. prophase I of meiosis;
2. mitotic metaphase    6. anaphase I of meiosis;
3. mitotic anaphase    7. metaphase II of meiosis;
4. mitotic telophase    8. anaphase II of meiosis.

**105. The following biopolymers: nucleic acids, proteins and polysaccharides are characterized by the following features:**

**The codes of the answers:**

1. linear, almost never branched;
2. linear or branched;
3. homopolymers;
4. heteropolymers.

**Put the correct answers into the table using the respective letters.**

| № | biopolymers     | characteristics |
|---|-----------------|-----------------|
| 1 | nucleic acids   |                 |
| 2 | proteins        |                 |
| 3 | polysaccharides |                 |

**106. Which one of the following statements is correct?**

**The codes of the answers:**

- protein function*
1. antibodies bind to parts of the antigen;
  2. antibodies fix oxygen;
  3. hemoglobin binds to oxygen;
  4. hemoglobin binds to parts of the antigen.

Answers:

- a) 1 and 4
- b) 1 and 3
- c) 2 and 4
- d) 2 and 3;

**107. The following statements refer to the female reproductive system.**

**Select the correct statements:**

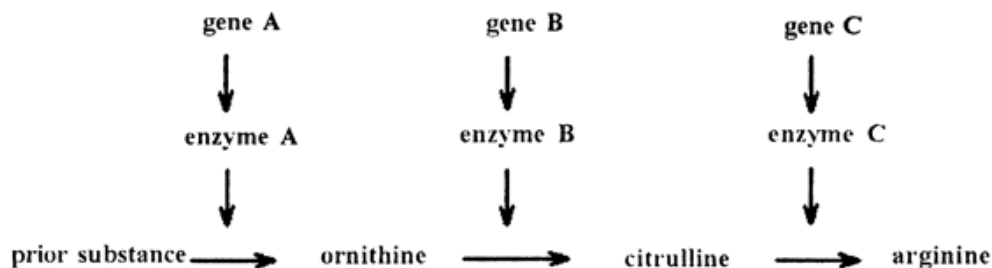
- a) both estrogen and progesterone are necessary for the ovulation to take place;
- b) estrogen tends to inhibit the production of FSH by the anterior pituitary gland;
- c) fertilization of the ovum by the sperm normally takes place in the uterus;
- d) progesterone production is largely under the control of LH;
- e) through out the part of the menstrual cycle that follows ovulation, there is a slight rise of body temperature.

\*\*\* Following information deals with the question № 108-113.

**The bread mould (*Neurospora*) normally produces its own amino acids from raw materials through a system of enzymes.**

**Use the following key for the items № 108-113:**

- a) a logical hypothesis based on the diagram;
- b) an illogical hypothesis;
- c) a hypothesis unrelated to the diagram;
- d) a restatement of information given by the diagram.



**108. Enzyme A catalyzes the reaction in which ornithine is formed from the prior substance.**

Answer:

a, b, c, d.

**109. If gene B were not present, arginine would be formed directly from ornithine.**

Answer:

a, b, c, d.

**110. If gene A were destroyed, the mould would survive if ornithine were added to the medium.**

Answer:

a, b, c, d.

**111. If gene B were destroyed, the mould could survive if enzyme B were added to the medium.**

Answer:

a, b, c, d.

**112. Gene C directs the formation of enzyme C** Answer:

a, b, c, d.

**113. Different moulds have different amino acid requirements.**

Answer:

a, b, c, d.

*\*\*\* Following information deals with the question № 114-115.*

**In a random-mating population of plant A (a cross-fertilizing plant), the following phenotypic frequencies are found:**

**Tall, green stem - 63%;**

**Tall, purple stem - 21%;**

**Short, green stem - 12%;**

**Short, purple stem - 4%.**

**T - a dominant allele controlling tall stem;**

**t - a recessive allele controlling short stem;**

**R - a dominant allele controlling green stem;**

**r - a recessive allele controlling purple stem.**

**114. What is the percentage of the plants having the TT genotype?**

a) 84;

b) 63

c) 36;

d) 21; e) 12 f) 9 g) 4.

**115. What is the percentage of the plants having the TTRR genotype?**

a) 84;

b) 63

c) 36;

d) 21; e) 12 f) 9 g) 4.

*\*\*\* Following information deals with the question № 116-120.*

**a) gene flow;**

**b) sexually reproducing species;**

**c) natural selection;**

**d) mutation;**

**e) genetic drift.**

**Use the following key for the phrases.**

**116. The sole source of new alleles is:**

**117. Change of increase or decrease in relative allele frequencies of a population is:**

**118. One or more populations, the members of which interbreed and produce fertile offspring is:**

**119. Change in allele frequencies in a population due to immigration, emigration is:**

**120. Differential survival and reproduction of variant members of a population is:**